Data Mining

Data mining –Unit2

Unit II

Getting to Know about your Data-data objects and attribute types, basic statistical description of data, measuring data similarity and dissimilarity, Data Pre-processing: Cleaning– Integration–Reduction–Data Transformation

DATA PREPROCESSING

It refers to techniques and procedure used to prepare raw data into clean, organized structure format suitable for analysis or modelling

- Improves data Quality
- Enhance model performance
- Reducing Computational Complexity
- Ensuring Computability

major steps involved in data preprocessing, namely, data cleaning, data integration, data reduction, and data transformation

DATA CLEANING

Data cleaning routines work to “clean” the data by filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies. If users believe the data are dirty, they are unlikely to trust the results of any data mining that has been applied. Furthermore, dirty data can cause confusion for the mining procedure, resulting in unreliable output.

DATA CLEANING

Missing Values

1. Ignore the tuple: This is usually done when the class label is missing (assuming the
mining task involves classification). This method is not very effective, unless the tuple contains several attributes with missing values

2.Fill in the missing value manually: In general, this approach is time consuming and
may not be feasible given a large data set with many missing values.

3. Use a global constant to fill in the missing value: Replace all missing attribute valuesby the same constant such as a label like *"Unknown"* or "infinity".

4.Use a measure of central tendency for the attribute (e.g., the mean or median) to
fill in the missing value: For normal (symmetric) data distributions ,the mean can be used, while skewed data distribution should employ the median. Use this value to replace the missing value for *income*

5. Use the attribute mean or median for all samples belonging to the same class as
the given tuple: if classifying customers according to *credit risk*, we may replace the missing value with the mean *income* value for customers in the same credit risk category as that of the given tuple

6. Use the most probable value to fill in the missing value: This may be determined

II)Noisy Data

Noise is a random error or variance in a measured variable.

Binning: Binning methods smooth a sorted data value by consulting "neighborhood,"that is, the values around it. The sorted values are distributed into a number of "buckets," or *bins*.

In this example, the data for *price* are first sorted and then partitioned into *equal-frequency* bins of size 3 (i.e., each bin contains three values).

(1)**smoothing by bin means**, each value in a bin is replaced by the mean value of the bin. For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9.

(2) **smoothing by bin medians** can be employed, in which each bin value
is replaced by the bin median

(3)smoothing by bin boundaries, the minimum and maximum values in a given bin are identified as the *bin boundaries*. Each bin value is then replaced by the closest boundary value.

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:

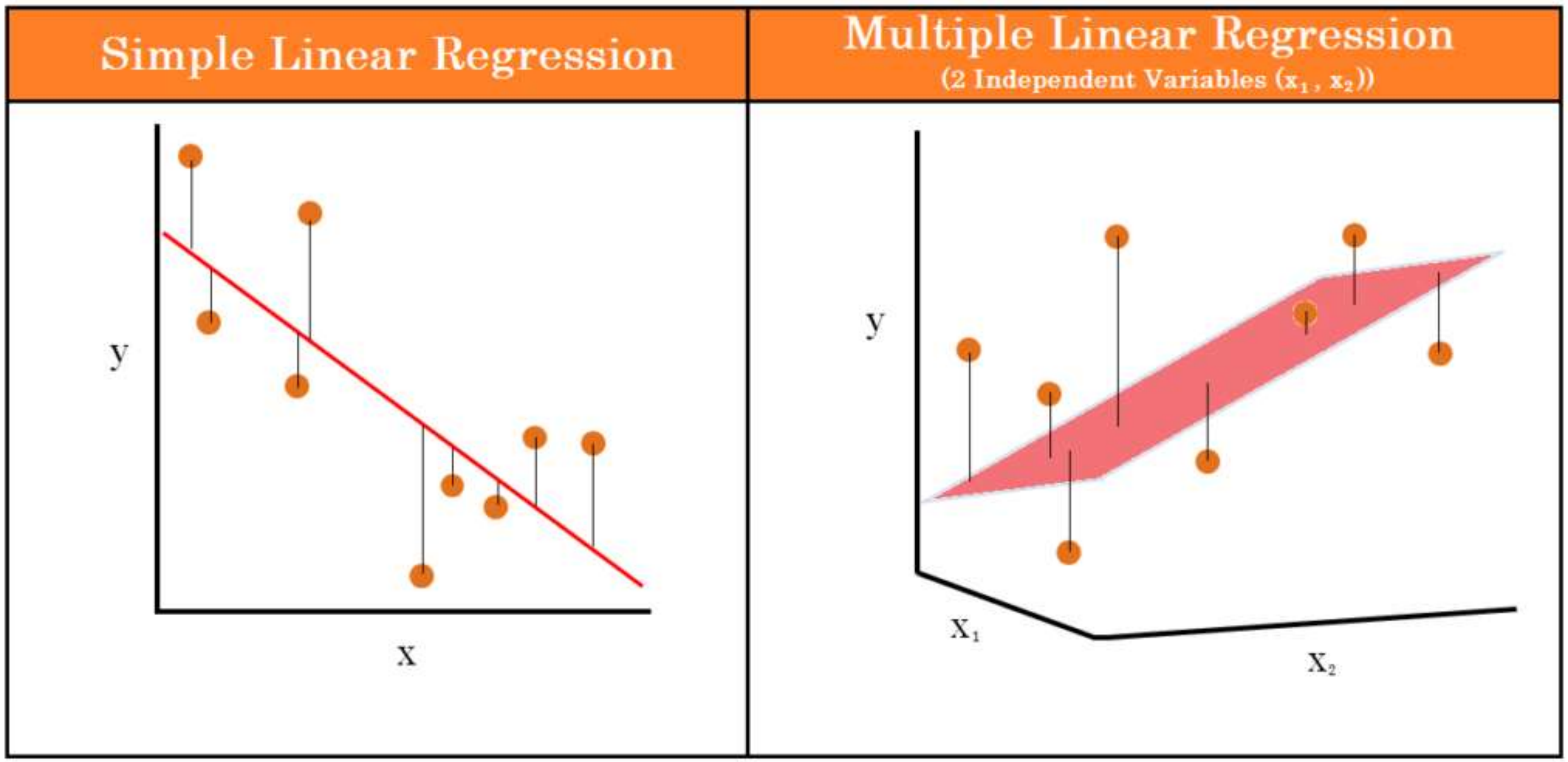
Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

: **3.2** Binning methods for data smoothing.

(**iii) Regression:** Data smoothing can also be done by regression, a technique that conforms
data values to a function.
Regression in data smoothing is a method where a best-fit line or curve is drawn through noisy data points to reduce fluctuations and show the overall trend.
Linear regression involves finding the "best" line to fit two attributes (or variables) so that one attribute can be used to predict the other.
Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface



Simple Linear Regression
y
x
Multiple Linear Regression
(2 Independent Variables (x_1, x_2))
y
X_1
X_2

(iv) **Outlier analysis**: Outliers may be detected by clustering, for example, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered outliers

3.2 Data Cleaning **91**

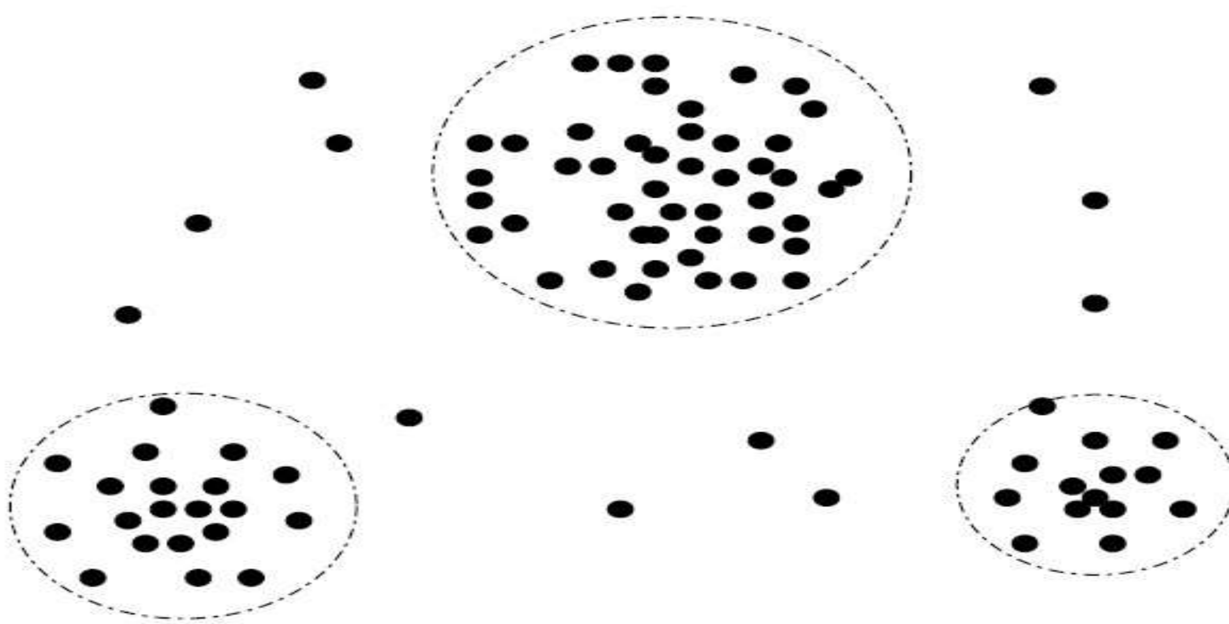


Figure 3.3 A 2-D customer data plot with respect to customer locations in a city, showing three data clusters. Outliers may be detected as values that fall outside of the cluster sets.

Major Tasks in Data Preprocessing

data integration

- This would involve integrating multiple databases, data cubes, or files.

different names in different databases, causing inconsistencies and redundancies.

- There are a number of issues to consider during data integration.
    - Schema integration
    - object matching.
- **Entity Identification Problem:** equivalent real-world entities from multiple data sources be matched up? This is referred to as the entity identification problem.
  - For example, how can the data analyst or the computer be sure that customer id in one database and cust number in another refer to the same attributes.
  - When matching attributes from one database to another during integration, special attention must be paid to the structure of the data.

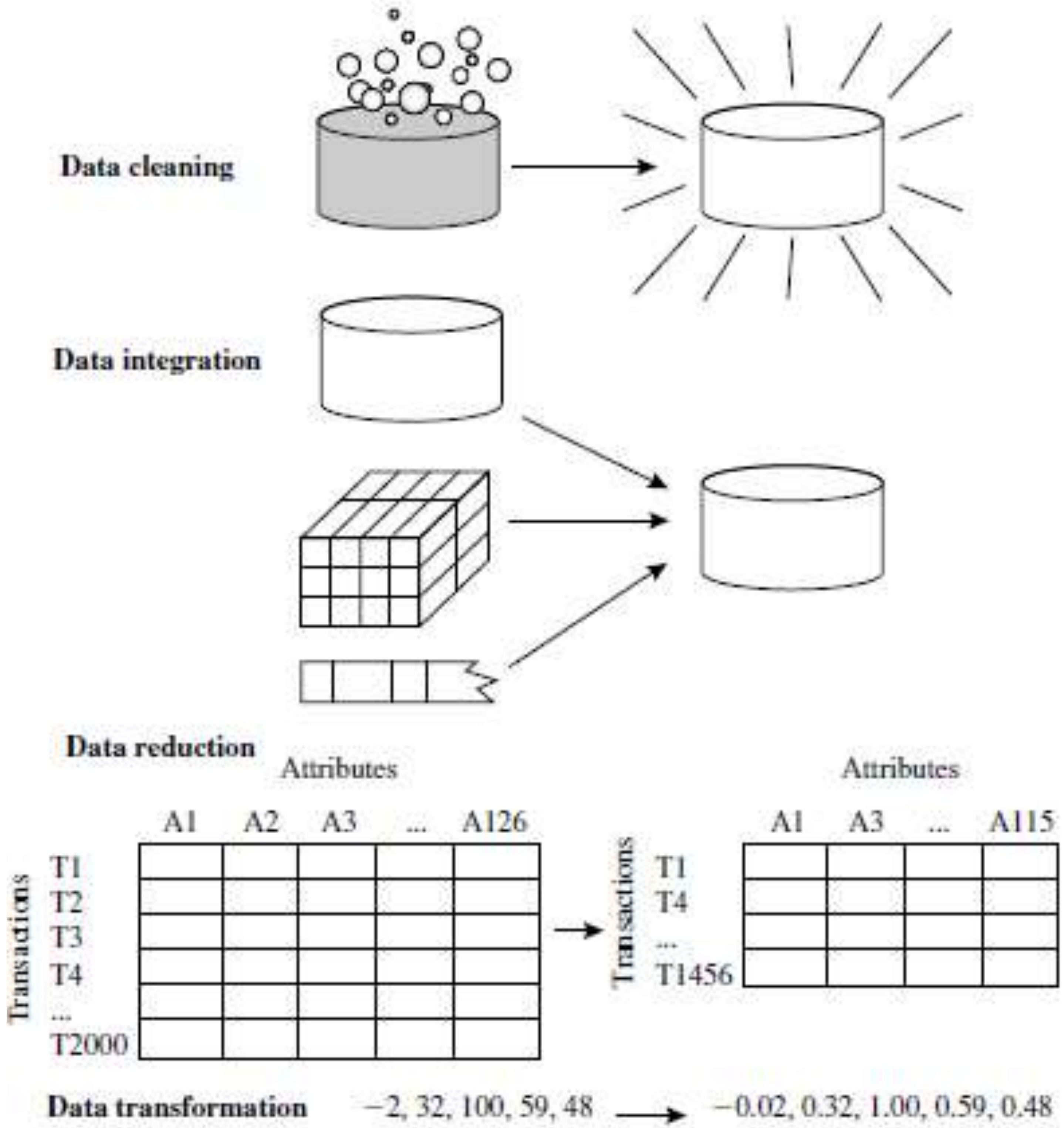
- **Redundancy and Correlation Analysis**
  - *Redundancy :* an attributes can be “derived” from another attribute or set of attributes.
  - Inconsistencies in attribute or dimension naming can also cause redundancies in the resulting data set.
  - Some redundancies can be detected by **correlation analysis**.
  - Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data.
  - For nominal data, we use the 2 (*chi-square*) test. For numeric attributes, we can use the *correlation coefficient* and *covariance*, both of which access how one attribute’s values vary from those of another.

Data reduction

- It obtains a reduced representation of the data set that is much smaller in volume, yet produces the same (or almost the same) analytical results.
- It include dimensionality reduction and numerosity reduction.
- dimensionality reduction, data encoding schemes are applied so as to obtain a reduced or "compressed" representation of the original data. (e.g., wavelet transforms and principal components analysis),
- Such methods provide better results if the data to be analyzed have been normalized, that is, scaled to a smaller range such as [0.0, 1.0].
- Normalization and concept hierarchy generation are forms of data transformation.
- You soon realize such data transformation operations are additional data preprocessing procedures that would contribute toward the success of the mining process.

Data Transformation

Data Transformation: It involves converting data into a format suitable for analysis.
Data Normalization: The process of scaling data to a common range to ensure consistency across variables.
Discretization: Converting continuous data into discrete categories for easier analysis.
Data Aggregation: Combining multiple data points into a summary form, such as averages or totals, to simplify analysis.
Concept Hierarchy Generation: Organizing data into a hierarchy of concepts to provide a higher-level view for better understanding and analysis.



Forms of data preprocessing.

Data Objects and Attribute Types

Data objects

- Data sets are made up of **data objects**.
- A data object represents an entity—in a sales database, the objects may be customers, store items, and sales
- Data objects are typically described by attribute
- If the data objects are stored in a database, they are data tuples.
- That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.

Attribute

- An attribute is a data field, representing a characteristic or feature of a data object. The nouns attribute, dimension, feature, and variable are often used interchangeably in the literature.
- The term dimension is commonly used in data warehousing.
- Machine learning literature tends to use the term feature, while statisticians prefer the term variable. Data mining and database professionals commonly use the term attribute
- The type of an attribute is determined by the set of possible values—**nominal, binary, ordinal, or numeric—the attribute can have**

Nominal Attributes

- Nominal means “relating to names.”
- The values of a nominal attribute are symbols or names of things.
- Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as categorical.
- The values do not have any meaningful order.
- Suppose that hair color and marital status are two attributes describing person objects.
- In our application, possible values for hair color are black, brown, blond, red, auburn, gray, and white.
- The attribute marital status can take on the values single, married, divorced, and widowed

Binary Attributes

- A binary attribute is a nominal attribute with only two categories or states**: 0 or 1**, where **0** typically means that **the attribute is absent**, and **1** means that it is **present.** Binary attributes are referred to as Boolean if the two states correspond to true and false.
- Given the attribute smoker describing a patient object, 1 indicates that the patient smokes, while 0 indicates that the patient does not.
- A binary attribute is **symmetric** if both of its states are equally valuable and carry the same weight; that is, there is no preference on which outcome should be coded as 0 or 1. One such example could be the attribute gender having the states male and female.
- A binary attribute is **asymmetric** if the outcomes of the states are not equally important, such as the positive and negative outcomes of a medical test for HIV.

Ordinal Attributes

- An ordinal attribute is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known Ordinal attributes.
- Suppose that drink size corresponds to the size of drinks available at a fast-food restaurant.
- This nominal attribute has three possible values: small, medium, and large. The values have a meaningful sequence (which corresponds to increasing drink size); however, we cannot tell from the values how much bigger, say, a medium is than a large.

Numeric Attributes

- A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values. Numeric attributes can be **interval-scaled or ratio-scaled**.

- **Interval-scaled attributes** are measured on a scale of equal-size units.
- The values of interval-scaled attributes have order and can be positive, 0, or negative.
- Thus, in addition to providing a ranking of values, such attributes allow us to compare and quantify the difference between values.
- temperature attribute is interval-scaled.
- Suppose that we have the outdoor temperature value for a number of different days, where each day is an object.
- By ordering the values, we obtain a ranking of the objects with respect to temperature. In addition, we can quantify the difference between values. For example, a temperature of 20C is five degrees higher than a temperature of 15C.
- Calendar dates are another example. For instance, the years 2002 and 2010 are eight years apart.

Ratio-Scaled Attributes

- A ratio-scaled attribute is a numeric attribute with an inherent zero-point. That is, the value as being a multiple (or ratio)of another value.
- In addition, the values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode.
- Unlike temperatures in Celsius and Fahrenheit, the Kelvin (K) temperature scale
- Other examples of ratio-scaled attributes include count attributes such as years of experience (e.g., the objects are employees) and number of words (e.g., the objects are documents).

Discrete versus Continuous Attributes

- A discrete attribute has a finite or countably infinite set of values, which may or may not be represented as integers.
- The attributes hair color, smoker, medical test, and drink size each have a finite number of values, and so are discrete.
- Note that discrete attributes may have numeric values, such as 0 and 1 for binary attributes or, the values 0 to 110 for the attribute age.
- An attribute is countably infinite if the set of possible values is infinite but the values can be put in a one-to-one correspondence with natural numbers.
- For example, the attribute customer ID is countably infinite. The number of customers can grow to infinity, but in reality, the actual set of values is countable
- If an attribute is not discrete, it is continuous..

7. Dynamic and Changing Data

In many applications, data is dynamic and constantly changing. This poses a challenge for data mining techniques that must adapt to new information and evolving patterns input data.

Solutions:

- Implement **real-time data mining** systems that can update and adapt to new data on-the-fly.
- Use **ad hoc data mining** approaches to handle unplanned and dynamic data queries.
- Develop **data mining approaches** that can learn incrementally and adjust to changing data landscapes.